VNR Vignana Jyothi Institute of Engineering

Civil Engineering | Course Code: CE301

Structural Analysis & Indeterminate Systems - Comprehensive Syllabus Lec

Syllabus Focus: CE301 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Structural Analysis & Indeterminate Systems.

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Structural Indeterminacy & Energy Methods

" Syllabus Scope: Static and kinematic indeterminacy of beams, frames, and trusses, strain energy, Castig

Unit 2: Slope-Deflection Method for Beams & Frames

" Syllabus Scope: Slope-deflection equations, continuous beams with support settlements, non-sway and swa

CE301: Structural Analysis & Indeterminate Systems - Advanced Topics & Assessment

Academic Track: Civil Engineering - Units 3, 4, 5 & Examination Matrix

---

II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Moment Distribution Method Hardy Cross

" Stiffness factors, carry-over factors, distribution factors, continuous beams with sinking supports, single-

Unit 4: Influence Lines for Moving Loads

" Influence line diagrams ILD for reactions, shear force, and bending moments in determinate beams, Muller-Bre

Unit 5: Matrix Methods of Structural Analysis

" Flexibility matrix method force method, Stiffness matrix method displacement method, element coordinate tran

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Analyze a continuous beam ABC having a settlement of 5 mm at support B using the Slope-Deflection Meth
2. Apply the Moment Distribution Method to analyze a two-span continuous beam with fixed end supports and
3. State Muller-Breslaus Principle and draw the Influence Line Diagram ILD for the bending moment at mid-

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* C.S. Reddy - Basic Structural Analysis McGraw-Hill 3rd Ed
* R.C. Hibbeler - Structural Analysis Pearson 9th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students